

Q.1 Determine the asymptotic complexity of $T(n)$.

(GATE 2021 Style)

$$T(n) = T(n/2) + T(2n/5) + 7n$$

$$T(0) = 1$$

Answer: $\Theta(n)$

Q2 Under what conditions will:

(GATE 2021 Style)

$$T(n) = \Theta(n \log n)$$

hold?

For constants $a \geq 1$ and $b > 1$:

$$T(n) = aT(n/b) + f(n)$$

Answer: $a = b$ and $f(n) = \Theta(n)$

Q.3 An algorithm checks whether an array of size N is sorted by making a single pass and comparing only adjacent elements.

What is the tightest asymptotic running time?

(GATE 2024 Style)

Answer: $\Theta(N)$

Q.4 Evaluate the asymptotic order of:

(GATE 2015 Style)

$$1^3 + 2^3 + 3^3 + \dots + n^3$$

Which of the following are true?

$$\Theta(n^4)$$

$$\Theta(n^5)$$

$$O(n^5)$$

$$\Omega(n^3)$$

Answer: $\Theta(n^4)$, $O(n^5)$, $\Omega(n^3)$

Q.5 Find the asymptotic value of q .

(GATE 2015 Style)

```
int fun1(int n)
```

```
{
```

```
    int i, j, k, p, q=0;
```

```
    for(i=1;i<n;++i)
```

```
    {
```

```
        p=0;
```

```
        for(j=n;j>1;j=j/2)
```

```
            ++p;
```

```
        for(k=1;k<p;k=k*2)
```

```
            ++q;
```

```

    }

    return q;
}
Answer:  $\Theta(n \log(\log n))$ 

```

Q.6 Determine the complexity: (Classic GATE)

```

for(int i=n;i>0;i/=2)
{
    for(int j=0;j<i;j++)
    {
        // constant work
    }
}

```

Answer: $\Theta(n)$

Q.7 Determine the complexity: (Classic GATE)

```

for(int i=1;i<=n;i++)
{
    for(int j=1;j<=i;j++)
    {
        // constant work
    }
}

```

Answer: $\Theta(n^2)$

Q.8 Determine the complexity: (Recursion)

```

void fun(int n)
{
    if(n<=1)
        return;

    fun(n/2);
    fun(n/2);
}

```

Answer: $\Theta(n)$

Q.9 Determine the complexity: (Fibonacci Pattern)

```

int fib(int n)
{
    if(n<=1)

```

```

    return n;

    return fib(n-1)+fib(n-2);
}

```

Answer: $\Theta(\phi^n) \approx O(2^n)$

Q.10 Arrange the following in increasing order:

(GATE Favorite)

```

n^(3/2)
n log n
n^(log2 n)
2^n

```

Answer: $n \log n < n^{(3/2)} < n^{(\log_2 n)} < 2^n$

Q.11 Determine:

$T(n)=T(n-1)+1$

Answer: $\Theta(n)$

Q.12 Determine:

$T(n)=T(n-1)+n$

Answer: $\Theta(n^2)$

Q.13 Determine:

$T(n)=2T(n-1)+1$

Answer: $\Theta(2^n)$

Q.14 Determine:

$T(n)=T(n/2)+1$

Answer: $\Theta(\log n)$

Q.15 Determine:

```

for(int i=1; i<=n; i*=2)
{
    for(int j=1; j<=n; j++)
    {
    }
}

```

Answer: $\Theta(n \log n)$